

Full Marks:50

SET:A

Time: 1Hour 10Minutes

Q1. What will be printed?

```
x = 10
```

```
def outer():
```

```
    x = 20
```

```
    def inner():
```

```
        print(x)
```

```
    x = 30
```

```
    inner()
```

```
outer()
```

```
A)20
```

```
B)10
```

```
C)30
```

```
D)NameError
```

Q2. What does the following function return?

```
def fun():
```

```
    print("Hello")
```

```
x = fun()
```

```
print(x)
```

```
A)Error
```

```
B)Hello None
```

```
C)None
```

```
D)Hello
```

Q3. What will be the output?

```
x = 10
```

```
def outer():
```

```
    x = 20
```

```
    def inner():
```

```
        global x
```

```
        x = 30
```

```
inner()
```

```
print(x)
```

```
outer()
```

```
print(x)
```

```
A)30
```

```
30
```

```
B)20
```

```
10
```

```
C)20
```

```
30
```

```
D)30
```

```
20
```

Q4. What is the output?

```
try:
```

```
    x = int("abc")
```

```
except ValueError:
```

```
    print("A")
```

```
else:
```

```
    print("B")
```

```
finally:
```

```
    print("C")
```

```
A)B
```

```
B)A
```

```
C)A C
```

```
D)B C
```

Q5. Which object keeps track of modules that have already been imported during the current Python process?

```
A)__module__
```

```
B)sys.imports
```

```
C)sys.modules
```

```
D)sys.path
```

Q6. What happens here?

```
class MyError(Exception):
```

```
    pass
```

```
def test():
```

```
    raise MyError("Something went
```



wrong")

try:

test()

except MyError as e:

print(type(e).__name__)

A)TypeError

B)MyError

C)Something went wrong

D)Exception

Q7. What is the output?

def fun(a=10, b=20):

return a - b

print(fun(30))

A)-10

B)20

C)10

D)30

Q8. What is the output?

try:

raise ValueError

except ValueError:

print("A")

raise

A)No Output

B)A, followed by propagation of the same exception

C)A, and execution ends normally

D)B

Q9. What will happen?

def fun(a, b=10):

return a + b

print(fun(b=5, a=20))

A)SyntaxError

B)TypeError

C)25

D)15

Q10. What is the output?

def fun(x):

return x * 2

a = fun

print(a(5))

A)10

B)func(5)

C>Error

D)5

Q11. What will normally happen?

import math

import math

A)ImportError

B)The module is normally loaded and executed twice

C)The module is normally loaded once and reused

D)SyntaxError

Q12. What is the output?

def fun():

try:

print("A")

raise ValueError

except ValueError:

print("B")

return 10

else:

print("C")

return 20

finally:

print("D")

print(fun())

A)A

B

C

D

10

B)A

B

D

10

C)A

C

D

20

D)A

B

10



D

Q13. What will be printed?

```
def fun():  
    print("A")  
    return  
    print("B")
```

fun()

A)A B

B)B

C)A

D)Nothing

Q14. Why is

try:

```
    print(10 / 0)
```

except Exception:

```
    print("A")
```

except ZeroDivisionError:

```
    print("B")
```

codes's second handler never reached?

A)Exception is a superclass of ZeroDivisionError

B)ZeroDivisionError occurs before Exception

C)ZeroDivisionError is not an exception

D)except blocks execute randomly

Q15. if test.py has

```
print("A")
```

```
if __name__ == "__main__":  
    print("B")
```

If test.py is executed directly, what will be printed?

A)Nothing

B)A followed by B

C)B

D)A

Q16. What is the output?

```
def fun(x):  
    if x == 0:  
        return  
    return x + fun(x - 1)
```

print(fun(3))

A)3

B)None

C)5

D)6

Q17. What will be printed?

```
def fun(x=[]):  
    x.append(10)  
    return x
```

a = fun()

b = fun()

print(a)

print(b)

A)[10]

[10, 10]

B)[10]

[10]

C)Error

D)[10, 10]

[10]

Q18. What is the output?

def outer():

```
    x = 10
```

def inner():

```
    nonlocal x
```

```
    x = x + 5
```

```
    return x
```

inner()

```
    return x
```

print(outer())

A)5

B)UnboundLocalError

C)15

D)10

Q19. What happens when this function is called?

def fun(a=10, b):

```
    return a + b
```

A)NameError

B)SyntaxError

C)TypeError

D)It executes correctly



Q20. What happens here?

```
def fun(a):  
    return a
```

```
print(fun())
```

- A)None
- B)NameError
- C)0
- D)TypeError

Q21. Suppose the current directory contains a file named random.py. What potential problem can occur with:

```
import random
```

- A)random cannot be imported
- B)The local file can shadow the intended standard-library module
- C)Python automatically renames the local file
- D)Python will always reject it

Q22. What happens here?

```
from math import sqrt
```

```
print(sqrt(36))
```

- A)6
- B)36
- C)NameError
- D)6.0

Q23. What is the output?

```
def fun():  
    try:  
        return 10  
    finally:  
        return 20
```

```
print(fun())
```

- A)10
- B)20
- C)None
- D)RuntimeError

Q24. Suppose module_a.py contains:

```
print("Module loaded")
```

value = 10

And another file contains:

```
import module_a  
import module_a
```

```
print(module_a.value)
```

Assuming both imports occur in the same Python process, what is normally printed?

- A)Module loaded
- 10
- B)Module loaded
- Module loaded
- 10
- C)ImportError
- D)10

10

Q25. What is the output?

```
numbers = [1, 2, 3, 4, 5]
```

```
result = list(  
    map(  
        lambda x: x * 2,  
        filter(lambda x: x % 2 == 0,  
            numbers)  
    )  
)
```

```
print(result)
```

- A)[2, 4, 6, 8, 10]
- B)[4, 8]
- C)[1, 3, 5]
- D)[2, 4]

Q26. What will be the output?

```
x = 10
```

```
def fun():  
    print(x)  
    x = 20
```

```
fun()
```

- A)10
- B)UnboundLocalError



C)20

D)NameError

Q27. What will be printed?

try:

`print(10 / 0)`

except ZeroDivisionError:

`print("A")`

except Exception:

`print("B")`

A)A

B)A B

C)Nothing

D)B

Q28. What will happen?

def fun(n):

`if n == 0:`

`return 1`

`return n * fun(n)`

print(fun(3))

A)TypeError

B)6

C)3

D)RecursionError

Q29. What happens here?

try:

`raise ValueError("Wrong value")`

except ValueError as e:

`print(e)`

A)Wrong Value

B)ValueError

C)e

D)Nothing

Q30. What will be the output?

import math

print(math.sqrt(25))

A)sqrt(25)

B)5.0

C)5

D)25

Q31. What will be printed?

try:

`x = int("10")`

except ValueError:

`print("A")`

else:

`print("B")`

finally:

`print("C")`

A)A

B)B C

C)A C

D)B

Q32. What will be printed?

def fun():

try:

`return 10`

finally:

`print("Finally")`

print(fun())

A)Finally followed by 10

B)10 followed by Finally

C)10 only

D)Finally only

Q33. What will be printed?

from functools import reduce

numbers = [1, 2, 3, 4]

result = reduce(

`lambda a, b: a * b,`

`numbers`

)

print(result)

A)0

B)10

C)9

D)24

Q34. What is the output?

def outer():

`x = 10`

def inner():

`return x + 5`



return inner

```
f = outer()
print(f())
```

A)NameError
B)15
C)None
D)10

Q35. Suppose sample.py contains:

```
x = 10
```

```
def display():
    print("Hello")
```

What will this print?

```
import sample
```

```
print(sample.x)
sample.display()
```

A)10
Hello
B)x
display
C)Error
D)Only Hello

Q36. What happens?

```
def fun(a, b):
    return a + b
```

```
print(fun(10, a=20))
```

A)SyntaxError
B)20
C)TypeError
D)30

Q37. What is the output?

```
f = lambda x: x + 5
```

```
print(f(10))
```

A)15
B)Error
C)10
D)5

Q38. What happens when this code runs?

```
def first():
    second()
```

```
def second():
    third()
```

```
def third():
    raise RuntimeError("Failure")
```

```
try:
    first()
except RuntimeError:
    print("Handled")
```

A)TypeError
B)RuntimeError cannot propagate through functions
C)Handled
D)NameError

Q39. What will happen?

```
def outer():
    x = 10
```

```
    def inner():
        nonlocal y
        y = 20
```

```
    inner()
```

```
outer()
```

A)20
B)NameError
C)10
D)SyntaxError

Q40. What is the output?

```
def fun(x):
    x += 10
    return x
```

```
x = 5
y = fun(x)
```

```
print(x, y)
```

A)15 15
B)5 15
C)15 5



D)5 5

Q41. What will be printed?

```
x = 10
```

```
f = lambda: x
```

```
x = 20
```

```
print(f())
```

A)None

B)NameError

C)10

D)20

Q42. What will be printed?

```
def fun(a, b):  
    return a + b
```

```
print(fun(10, b=20))
```

A)SyntaxError

B)30

C)TypeError

D)20

Q43. What happens when this code executes?

```
def fun():  
    print(variable)  
    variable = 50
```

```
variable = 10
```

```
fun()
```

A)UnboundLocalError

B)NameError

C)50

D)10

Q44. What is the major difference between the following?

```
import math
```

```
and
```

```
from math import sqrt
```

A)First imports only sqrt

B)Both import only sqrt

C)First provides the module namespace;
second imports sqrt directly

D)There is no difference

Q45. Suppose test.py contains:

```
print("A")
```

```
if __name__ == "__main__":  
    print("B")
```

If another Python program executes:

```
import test
```

what will normally be printed?

A)Nothing

B)A and B

C)B

D)A

Q46. What is the output?

```
try:  
    print(10 / 0)  
except Exception:  
    print("A")  
except ZeroDivisionError:  
    print("B")
```

A)SyntaxError

B)A

C)B

D)A B

Q47. What will be the output?

```
def fun(x):  
    try:  
        if x < 0:  
            raise ValueError("negative")  
        return x  
    except ValueError:  
        return 100  
    finally:  
        print("cleanup")
```

```
print(fun(-5))
```

A)negative

cleanup

B)100

cleanup



C)cleanup

100

D)cleanup

-5

Q48. What will be printed?

```
def make_counter():  
    count = 0
```

```
    def counter():  
        nonlocal count  
        count += 1  
        return count
```

```
    return counter
```

```
a = make_counter()  
b = make_counter()
```

```
print(a())  
print(a())  
print(b())  
print(a())
```

A)1

2

1

3

B)1

2

3

4

C)0

1

0

2

D)1

1

1

1

Q49. What is the purpose of this statement?

```
if __name__ == "__main__":
```

A)It prevents all imports

B)It creates the main function

C)It identifies code intended to run when the file is executed directly

D)It prevents syntax errors

Q50. What will be printed?

```
class InvalidAgeError(Exception):  
    pass
```

```
def check_age(age):  
    if age < 18:  
        raise InvalidAgeError("Age is too low")  
    return "Allowed"
```

```
try:  
    print(check_age(15))  
except InvalidAgeError as e:  
    print(type(e).__name__)  
    print(e)
```

A)InvalidAgeError

15

B)Exception

Age is too low

C)Nothing, because custom exceptions cannot be caught

D)InvalidAgeError

Age is too low

